ESPIRA: Feasibility of Electrostatic Debris Removal in Low Earth Orbit

A conceptual design study and first-order physics assessment

Harshita Pandita

Independent project · November 2026 · An interactive 3D model, feasibility calculator and analysis code accompany this paper

ABSTRACT

ESPIRA (Electrostatic Space Particle Interception and Removal Architecture) is a concept for a ~380 kg spacecraft that charges orbital debris with an electron beam and UV light, then moves it with ±20 kV electrostatic panels. A second-generation design added five subsystems, including a 200 m charged filament web. This paper tests the core mechanism with first-order models of plasma screening, spacecraft charging and Coulomb force. In vacuum, the force on a 0.5 m object at 10 m is about 0.2 mN, which gives a 100 kg object roughly 0.2 m/s of Δv per day. At 400–550 km altitude, however, the ionospheric plasma forms a 15–180 m sheath around a 20 kV body. Holding that potential would require an estimated 19–24 kW, more than ten times the 1.4 kW array. Debris crossing at ~10 km/s carries about 10^6 times more kinetic energy than the field can remove. The concept is therefore not viable in low Earth orbit (LEO). Electrostatic actuation remains credible for slow co-orbital tasks in geostationary orbit (GEO), while in LEO the same plasma enables electrodynamic tethers. Two subsystems, an adaptive-adhesion capture net and tumble-aware capture logic, carry over to a revised concept.

1 Motivation

About 41,700 objects are regularly tracked in Earth orbit, of which about 11,600 are working satellites. ESA's MASTER-8 model estimates about 54,000 objects larger than 10 cm, 1.2 million between 1 and 10 cm, and 140 million between 1 mm and 1 cm [1]. ESA's 2025 Space Environment Report concludes that new debris is being added faster than old debris re-enters, and that active debris removal is needed to prevent a collision cascade [1].

Most removal concepts capture debris physically with nets, harpoons or robotic arms, which requires matching the motion of a tumbling, uncooperative object. ESPIRA explores a contactless alternative: charging debris and moving it with electric fields, as industrial electrostatic precipitators do with dust. This paper asks where that idea holds up.

2 The ESPIRA concept

All values in this section are design targets of the concept study and have not been tested. The baseline (V1) spacecraft has a ~380 kg dry mass, operates in a 550 km sun-synchronous orbit and carries a 1.4 kW solar array. It charges targets with an electron beam gun (0.5–5 keV, 0.5–8 m range) and a deep-UV emitter, and acts on them with four ±20 kV field panels. Supporting systems are an optical, LIDAR and radar sensor head, a three-joint arm with a 3 × 3 m net for objects larger than 10 cm, a storage bay and a xenon Hall thruster. The intended capture radius is about 10 m. Table 1 lists the five additions proposed for a second generation (V2).

Table 1. Proposed V2 subsystems.

Subsystem	Concept	Inspiration	Target
MYCENET	Nine charged 50 µm carbon-nanotube filaments	Slime-mould (*Physarum*) networks	200 m reach
SPINDAR	Nitrogen-vacancy diamond magnetometer for material ID	Quantum sensing	5–15 m range
ELECTROSKIN	Net tiles with voltage-switchable adhesion	Electrowetting microfluidics	Fewer bounce-offs
NEUROTRIAGE	Spiking neural network co-processor	Biological neurons	Low-energy decisions
STIGMA	Coordination through charge marks on debris	Ant stigmergy	No radio links

3 Method

The assessment uses standard first-order models, the same simplifications as the electrostatic-tractor literature [2,3]. They test orders of magnitude and do not replace a spacecraft-charging code such as SPIS or NASCAP-2k. The plasma screens a charge over the electron Debye length:

$$\lambda_D = (\varepsilon_0 \, k \, T_e / n_e \, e^2)^{1/2} \tag{1}$$

The probe (R = 1 m) and a debris object (R = 0.5 m) are treated as spheres with charge $Q = 4\pi\varepsilon_0 RV$ at ±20 kV. The force between them is the screened Coulomb force:

$$F = Q_1 Q_2 / (4\pi\varepsilon_0 d^2) \cdot (1 + d/\lambda_D) \cdot e^{-d/\lambda_D} \tag{2}$$

At kilovolt potentials in LEO, the screened region is a sheath far thicker than λ_D. Its thickness is estimated with the planar Child–Langmuir law, and the force inside it is taken as unscreened. The power to hold the potential is the plasma current collected across the sheath (ram ion flux for negative bias, electron thermal flux for positive bias), multiplied by the voltage. Plasma inputs are typical ranges: $n_e = 10^{10}$–10^{12} m^{-3} and T_e = 0.1–0.2 eV in LEO, and $n_e = 10^5$–10^6 m^{-3} and T_e = 1–10 keV in GEO. MYCENET encounter rates are computed as spatial density × web cross-section × 10 km/s relative speed.

4 Results

Table 2. Key results for the ±20 kV system.

Quantity	LEO, 400–550 km	GEO
Debye length	3–24 mm	~235 m (quiet) to ~2.4 km (storm)
Sheath around 20 kV probe	~15 m (day) to ~180 m (night)	Not limiting at tens of metres
Force on 0.5 m debris at 10 m	~0.22 mN (inside sheath)	~0.22 mN
Power to hold −20 kV	~19–24 kW (+20 kV: ~670–750 kW)	~0.2 W

Power, not range, is the binding limit in LEO. Inside the sheath the field is nearly unscreened, so the force at 10 m matches the vacuum value. Maintaining the voltage, however, requires collecting plasma current at a rate

that needs more than ten times the available array power (Fig. 1).

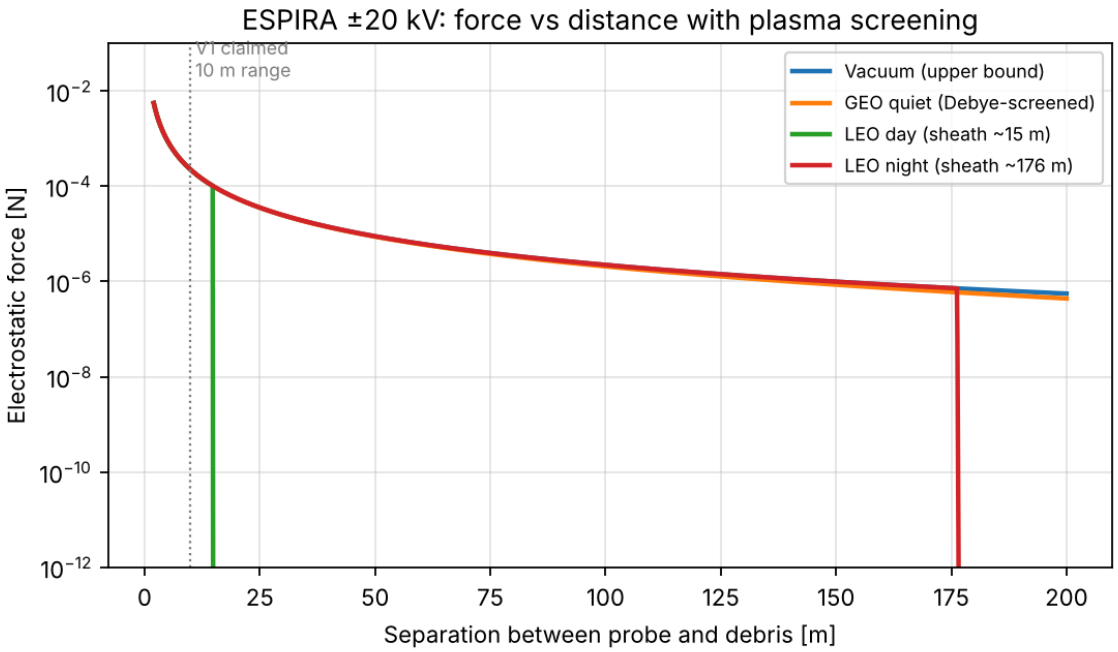


Figure 1. Electrostatic force against separation for the ±20 kV system. In LEO the force collapses at the sheath edge (~15 m by day, ~180 m by night); in GEO it follows the vacuum curve.

Force is too small for removal. Even in the best case (vacuum, 10 m), a month of continuous push changes a 100 kg object's speed by about 6 m/s. A GEO graveyard re-orbit (+300 km) needs about 11 m/s, while lowering a 550 km orbit to a fast-decay perigee needs about 100 m/s (Fig. 2).

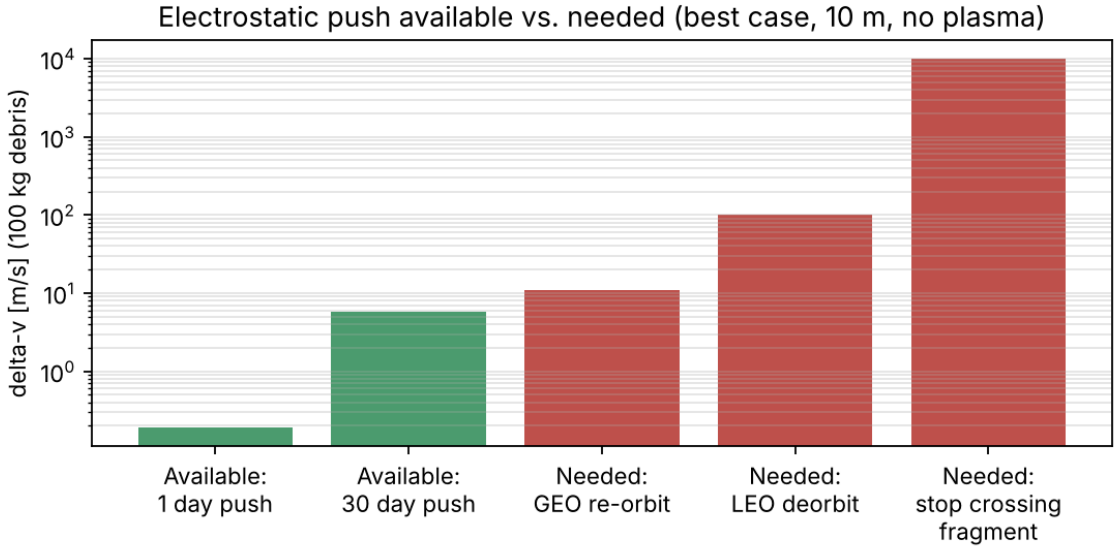


Figure 2. Δv available from continuous electrostatic push (100 kg object, best case) against Δv required for removal tasks. Log scale.

MYCENET encounters are rare and destructive. With realistic spatial densities (~10^{-7} km^{-3} for objects larger than 10 cm), the 9 × 200 m web would meet such an object about once every 175 years, and a 1–10 cm object about once every 20 years. The original design assumed a density about 10^{4} times higher. A 1 cm alu-

minium fragment crossing at 10 km/s carries ~71 kJ, while the field can do ~0.04 J of work on it over 200 m. The fragment would cut the filament rather than be captured.

5 Subsystem assessment

Table 3. Verdict on each V2 subsystem.

Subsystem	Verdict	Main reason
MYCENET	Not viable	Encounters are rare, and crossing fragments sever filaments.
SPINDAR	Doubtful	Signal from weakly magnetic materials falls roughly as $1/r^3$, so identification at metres is not established.
ELECTROSKIN	Worth testing	Works at contact range, so it is not limited by screening. Survival in vacuum and UV is the open question.
NEUROTRIAGE	Minor gain	Computing is a small share of the power budget.
STIGMA	Not viable	Charge marks are neutralised by ambient plasma current (~1 mA on a 0.5 m object) in well under a second.

Probe-level figures in earlier ESPIRA material (an annual clearance rate, an 8,000× coverage gain and lab-measured filament charge retention) were design assumptions, not results, and are withdrawn.

6 Revised direction

GEO: electrostatics. In GEO the Debye length is hundreds of metres and holding high voltage costs watts. A servicer can charge a target with an electron beam and tow it at millinewton forces. This is the electrostatic tractor, studied since 2011 [2,3], and the forces estimated here agree with its published values. ESPIRA's electron beam and field panels map onto this concept directly.

LEO: use the plasma. A bare conductive tether a few kilometres long collects electrons from the ionosphere. The current crossing Earth's magnetic field produces a Lorentz drag force with no propellant. As an order of magnitude, $F = ILB \approx 1\ \text{A} \times 1\ \text{km} \times 30\ \mu\text{T} \approx 30\ \text{mN}$, more than 100 times ESPIRA's best electrostatic force. Deorbit kits of this type were developed in the EU-funded E.T.PACK project [4,5]. ESPIRA's adaptive-grip net and tumble-aware capture logic could support a chaser that attaches such a kit to a large object, for example a spent rocket stage.

7 Limitations and next steps

The models rule out the original LEO concept but are too coarse for new design work. Isolated spheres stand in for real geometry. The sheath and current estimates use planar Child–Langmuir theory. Debris densities are order-of-magnitude values rather than ESA MASTER flux for a chosen orbit. Next steps are:

1. Reproduce a published electrostatic-tractor result in GEO with the multi-sphere method, to validate the model.
2. Model the deorbit time of an electrodynamic tether kit on a ~1 t object from 550–800 km, compared with a chemical deorbit.
3. Simulate contact between a tumbling target and passive versus adaptive-grip nets, to test whether ELECTROSKIN reduces bounce-off.

References

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3. J. Hammerl, H. Schaub, "Effects of electric potential uncertainty on electrostatic tractor relative motion control," *8th European Conference on Space Debris*, ESA.
4. G. Sarego, L. Olivieri, A. Valmorbida et al., "Deorbiting performance of electrodynamic tethers to mitigate space debris," University of Padova (E.T.PACK, H2020 grant 828902).
5. L. Olivieri, A. Valmorbida, G. Sarego et al., "Test of tethered deorbiting of space debris," *Advances in Astronautics Science and Technology*, 2020. doi:10.1007/s42423-020-00068-9

Analysis code: `analysis/espira_feasibility.py` reproduces every number in this paper.